

DIAGRAM CHASES: EXAMPLES & EXERCISES

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1. EXAMPLES OF DIAGRAM CHASES

1.1. Notation. Throughout, we write R for a commutative ring and $\mathbf{Mod}_R$ for the category of R -modules.

1.2. Lemma (5-lemma). *Consider a commutative diagram*

$$\begin{array}{ccccccccc}
 M_1 & \xrightarrow{\alpha_1} & M_2 & \xrightarrow{\alpha_2} & M_3 & \xrightarrow{\alpha_3} & M_4 & \xrightarrow{\alpha_4} & M_5 \\
 \gamma_1 \downarrow & & \downarrow \gamma_2 & & \downarrow \gamma_3 & & \downarrow \gamma_4 & & \downarrow \gamma_5 \\
 M'_1 & \xrightarrow{\alpha'_1} & M'_2 & \xrightarrow{\alpha'_2} & M'_3 & \xrightarrow{\alpha'_3} & M'_4 & \xrightarrow{\alpha'_4} & M'_5.
 \end{array} \tag{1.2.1}$$

in $\mathbf{Mod}_R$. If the rows are exact, then the following statements hold.

(1.2.a) If γ_2 and γ_4 are injective and γ_1 is surjective, then γ_3 is injective.

(1.2.b) If γ_2 and γ_4 are surjective and γ_5 is injective, then γ_3 is surjective.

Proof. First we prove (1.2.a). So suppose that γ_2 and γ_4 are injective and γ_1 is surjective. Start with $m_3 \in M_3$ with the property that $\gamma_3(m_3) = 0$. The goal is to show that $m_3 = 0$. Then since $\gamma_3(m_3) = 0$, we also know that $\alpha'_3 \gamma_3(m_3) = 0$, hence by the commutativity of the diagram (2.2.1) we have $\gamma_4 \alpha_3(m_3) = 0$. Since γ_4 is injective, this implies that $\alpha_3(m_3) = 0$, hence $m_3 \in \ker \alpha_3$. Since the top row is exact, $\ker \alpha_3 = \text{im } \alpha_2$, so there exists an $m_2 \in M_2$ such that $\alpha_2(m_2) = m_3$. Then since $\gamma_3(m_3) = 0$ we see that $\gamma_3 \alpha_2(m_2) = 0$. Then by the commutativity of the diagram (1.2.1) we see that $\alpha'_2 \gamma_2(m_2) = 0$. Therefore $\gamma_2(m_2) \in \ker \alpha'_2$.

Since the bottom row is exact, $\ker \alpha'_2 = \text{im } \alpha'_1$, so there exists an $m'_1 \in M'_1$ so that $\alpha'_1(m'_1) = \gamma_2(m_2)$. Since γ_1 is surjective, there exists a $m_1 \in M_1$ so that $\gamma_1(m_1) = m'_1$. Then by the commutativity of the diagram (1.2.1), we have

$$\gamma_2 \alpha_1(m_1) = \alpha'_1 \gamma_1(m_1) = \alpha'_1(m'_1) = \gamma_2(m_2).$$

Thus by the injectivity of γ_2 we have $\alpha_1(m_1) = m_2$. This shows that

$$\alpha_2 \alpha_1(m_1) = \alpha_2(m_2) = m_3.$$

But by the exactness of the top row, we have $\alpha_2 \alpha_1(m_1) = 0$, so $m_3 = 0$, as desired. Hence $\ker \gamma_3 = 0$, so γ_3 is injective. A summary of this chase is displayed below.

$$\begin{array}{ccccccc}
 m_1 & \longrightarrow & m_2 & \longrightarrow & m_3 & \longrightarrow & \alpha_3(m_3) \\
 \downarrow & & \downarrow & & \downarrow & & \downarrow \\
 m'_1 & \longrightarrow & \gamma_2(m_2) & \longrightarrow & 0 & \longrightarrow & 0.
 \end{array}$$

Now we prove (1.2.b).[†] So suppose that γ_2 and γ_4 are surjective and γ_5 is injective. Start by considering an element $m'_3 \in M'_3$: the goal is to show that m'_3 lies in the image of γ_3 . Map m'_3 to $\alpha'_3(m'_3)$. Then by the surjectivity of γ_4 there exists an element $m_4 \in M_4$ so that $\gamma_4(m_4) = \alpha'_3(m'_3)$. Now consider $\alpha'_4(m_4)$. Notice that by the commutativity of the diagram (1.2.1), we have

$$\gamma_5 \alpha'_4(m_4) = \alpha'_4 \gamma_4(m_4) = \alpha'_4 \alpha'_3(m'_3).$$

Then by the exactness of the bottom row, $\alpha'_4 \alpha'_3 = 0$, so $\gamma_5 \alpha'_4(m_4) = 0$. Therefore, by the injectivity of γ_5 we have $\alpha'_4(m_4) = 0$, so $m_4 \in \ker \alpha'_4$. By the exactness of the top row, $\ker \alpha'_4 = \text{im } \alpha'_3$, so there exists $m_3 \in M_3$ so that $\alpha'_3(m_3) = m_4$.

Now consider $m'_3 - \gamma_3(m_3)$. Notice that by the commutativity of the diagram (1.2.1),

$$\begin{aligned} \alpha'_3(m'_3 - \gamma_3(m_3)) &= \alpha'_3(m'_3) - \alpha'_3 \gamma_3(m_3) = \alpha'_3(m'_3) - \gamma_4 \alpha'_3(m'_3) \\ &= \alpha'_3(m'_3) - \gamma_4(m_4) = \alpha'_3(m'_3) - \alpha'_3(m'_3) = 0. \end{aligned}$$

This shows that $m'_3 - \gamma_3(m_3)$ is in the kernel of α'_3 , which is equal to $\text{im } \alpha'_2$ by the exactness of the lower sequence. Hence there exists an m'_2 so that $\alpha'_2(m'_2) = m'_3 - \gamma_3(m_3)$. Then by the surjectivity of γ_2 there exists an $m_2 \in M_2$ so that $\gamma_2(m_2) = m'_2$. Now consider $m_3 + \alpha_2(m_2)$ and notice that by the commutativity of the diagram (1.2.1), we have

$$\begin{aligned} \gamma_3(m_3 + \alpha_2(m_2)) &= \gamma_3(m_3) + \gamma_3 \alpha_2(m_2) = \gamma_3(m_3) + \alpha'_2 \gamma_2(m_2) \\ &= \gamma_3(m_3) + (m'_3 - \gamma_3(m_3)) = m'_3. \end{aligned}$$

This shows that γ_3 is surjective, as desired. $\square$

Now we prepare to prove the Snake lemma (Proposition 1.5).

1.3. Lemma. *Let R be a commutative ring and*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \alpha \downarrow & & \downarrow \beta \\ A' & \xrightarrow{f'} & B' \end{array} \quad (1.3.1)$$

a commutative square in $\mathbf{Mod}_R$.

(1.3.a) *The restriction $f|_{\ker(\alpha)}: \ker(\alpha) \rightarrow \ker(\beta)$ is the unique R -module homomorphism making the square*

$$\begin{array}{ccc} \ker(\alpha) & \xrightarrow{f|_{\ker(\alpha)}} & \ker(\beta) \\ \iota_\alpha \downarrow & & \downarrow \iota_\beta \\ A & \xrightarrow{f} & B \end{array}$$

commute, where ι_α and ι_β denote the inclusions. Moreover, if f is an injection, then so is $f|_{\ker(\alpha)}$.

[†]Actually this follows immediately by duality since $\mathbf{Mod}_R$ is an abelian category and could have been proven in the setting of abelian categories — maybe we'll talk about these at some point.

(1.3.b) Dually, there exists a unique R -module homomorphism $\bar{f}: \text{coker}(\alpha) \rightarrow \text{coker}(\beta)$ making the square

$$\begin{array}{ccc} A' & \xrightarrow{f'} & B' \\ q_\alpha \downarrow & & \downarrow q_\beta \\ \text{coker}(\alpha) & \xrightarrow{\bar{f}'} & \text{coker}(\beta) \end{array}$$

commute, q_α and q_β denote the quotient maps. Moreover, if f is a surjection, then so is $\bar{f}'$.

Proof. Since (1.3.b) is dual to (1.3.a), it suffices to prove (1.3.a). This follows easily from the universal property of the kernel. Consider the composite $\beta f \iota_\alpha$. By the commutativity of the square (1.3.1) and the fact that ι_α is the inclusion of the kernel of α we have that

$$\beta f \iota_\alpha = f' \alpha \iota_\alpha = 0.$$

Thus by the universal property of $\ker(\beta)$, there exists a unique R -module homomorphism $\ker(\alpha) \rightarrow \ker(\beta)$ making the square

$$\begin{array}{ccc} \ker(\alpha) & \xrightarrow{\quad} & \ker(\beta) \\ \downarrow \iota_\alpha & & \downarrow \iota_\beta \\ A & \xrightarrow{f} & B \end{array}$$

commute. The fact that the unique morphism $\ker(\alpha) \rightarrow \ker(\beta)$ is the restriction $f|_{\ker(\alpha)}$ follows immediately from the definition of the restriction.

The last thing to check is that if f is an injection, then so is $f|_{\ker(\alpha)}$. To see this, notice that if f is an injection, then since ι_α is an inclusion, $f \iota_\alpha$ is also an injection, so the composite $\iota_\beta \circ f|_{\ker(\alpha)}$ is an injection, which implies that $f|_{\ker(\alpha)}$ is an injection. $\square$

1.4. Lemma. Let R be a commutative ring and consider the following diagram of R -modules

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ \alpha \downarrow & & \downarrow \beta & & \downarrow \gamma \\ A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' \end{array} \quad (1.4.1)$$

(1.4.a) If the top row of (1.4.1) is exact and f' is an injection, then the sequence

$$\ker(\alpha) \xrightarrow{f|_{\ker(\alpha)}} \ker(\beta) \xrightarrow{g|_{\ker(\beta)}} \ker(\gamma)$$

is exact.

(1.4.b) Dually, if the bottom row of (1.4.1) is exact and g is a surjection, then the sequence

$$\text{coker}(\alpha) \xrightarrow{\bar{f}'} \text{coker}(\beta) \xrightarrow{\bar{g}'} \text{coker}(\gamma)$$

is exact.

Proof. As usual, since (1.4.b) is dual to (1.4.a), it suffices to prove (1.4.a). The only place where we need to show exactness is at $\ker(\beta)$. First let us show that $\text{im}(f|_{\ker(\alpha)}) \subset \ker(g|_{\ker(\beta)})$.

So suppose that $b \in \ker(\beta)$ and $b = f(a)$ for some $a \in \ker(\alpha)$. Then since the top row of (1.4.1) we see that

$$g|_{\ker(\beta)}(b) = g(b) = gf(a) = 0.$$

Hence $b \in \ker(g|_{\ker(\beta)})$, so we see that $\text{im}(f|_{\ker(\alpha)}) \subset \ker(g|_{\ker(\beta)})$.

Now let us show that $\ker(g|_{\ker(\beta)}) \subset \text{im}(f|_{\ker(\alpha)})$. Suppose that $b \in \ker(g|_{\ker(\beta)})$. Then by the exactness of the top row of (1.4.1) we know that $b = f(a)$ for some $a \in A$. Hence it suffices to show that $a \in \ker(\alpha)$. Since $b \in \ker(\beta)$ we know that $\beta(b) = 0$. But by the commutativity of (1.4.1) we see that

$$\beta(b) = \beta f(a) = f' \alpha(a).$$

Since f' is injective, we see that $\alpha(a) = 0$, as desired. $\square$

1.5. Proposition (Snake Lemma). *Let R be a commutative ring and consider the following diagram of R -modules, where the rows are exact*

$$\begin{array}{ccccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C & \longrightarrow & 0 \\ \alpha \downarrow & & \downarrow \beta & & \downarrow \gamma & & \\ 0 \longrightarrow & A' & \xrightarrow{f'} & B' & \xrightarrow{g'} & C' & \longrightarrow 0 \end{array} \quad (1.5.1)$$

*There exists a **connecting homomorphism** $\partial: \ker(\gamma) \rightarrow \text{coker}(\alpha)$, and the sequence*

$$\ker(\alpha) \xrightarrow{f|_{\ker(\alpha)}} \ker(\beta) \xrightarrow{g|_{\ker(\beta)}} \ker(\gamma) \xrightarrow{\partial} \text{coker}(\alpha) \xrightarrow{\bar{f}'} \text{coker}(\beta) \xrightarrow{\bar{g}'} \text{coker}(\gamma)$$

is exact. Moreover, if f is injective, then so is $f|_{\ker(\alpha)}$. Dually, if g' is surjective, then so is $\bar{g}'$.

Proof. There are three things to do: construct the connecting homomorphism, show that the connecting homomorphism is well-defined, and, in light of Lemma 1.4, show that the resulting sequence is exact at $\ker(\gamma)$ and $\text{coker}(\alpha)$. First we construct the connecting homomorphism. We start with an element $c \in \ker(\gamma)$. Since g is surjective, we can choose an element $b \in B$ such that $g(b) = c$. By the commutativity of the right-hand square of (1.5.1) and the fact that $c \in \ker(\gamma)$,

$$g' \beta(b) = \gamma g(b) = \gamma(c) = 0,$$

so by the exactness of the lower sequence of (1.5.1) and the injectivity of f' , there exists a unique $a' \in A'$ with the property that $f'(a') = \beta(b)$. We define the connecting homomorphism $\partial: \ker(\gamma) \rightarrow \text{coker}(\alpha)$ by the assignment

$$c \longmapsto [a'],$$

where $[a']$ denotes the class of a' in $\text{coker}(\alpha)$.

Now let us show that ∂ is well-defined as a set-theoretic map. Suppose that $\tilde{b} \in B$ is another element such that $g(\tilde{b}) = c$, and let $\tilde{a}' \in A'$ be the unique element with the property that $f'(\tilde{a}') = \beta(\tilde{b})$. Then since g is an R -module homomorphism,

$$g(b - \tilde{b}) = g(b) - g(\tilde{b}) = c - c = 0,$$

so by the exactness of the top row of (1.5.1) there exists an element $a \in A$ with the property that $f(a) = b - \tilde{b}$. By the commutativity of the left-hand square of (1.5.1), we see that

$$f' \alpha(a) = \beta f(a) = \beta(b - \tilde{b}) = \beta(b) - \beta(\tilde{b}).$$

Since $f'(a' - \tilde{a}') = \beta(b) - \beta(\tilde{b})$ and f' is injective, we see that $\alpha(a) = a' - \tilde{a}'$, that is $a' = \tilde{a}' + \alpha(a)$. Thus in $\text{coker}(\alpha)$ we see that

$$[a'] = [\tilde{a}' + \alpha(a)] = [\tilde{a}'] ,$$

so the connecting homomorphism is actually well-defined.

Now we need to prove that ∂ is an R -module homomorphism, but this is easy since we now know that the value of $\partial(c)$ is independent of any choices we make. Suppose that we are given $r \in R$ and $c \in \ker(\gamma)$. Choose $b \in B$ so that $\beta(b) = c$. Then since β is an R -module homomorphism, we have that $\beta(rb) = rc$. Let $a' \in A'$ be the unique element so that $f'(a') = \beta(b)$. Then since f' and β are R -module homomorphisms

$$f'(ra') = rf'(a') = r\beta(b) = \beta(rb) ,$$

hence

$$\partial(rc) = [ra'] = r[a'] = r\partial(c) .$$

Similarly, suppose that $c_1, c_2 \in \ker(\gamma)$, and let $b_1, b_2 \in B$ be elements such that $g(b_1) = c_1$ and $g(b_2) = c_2$, and let $a'_1, a'_2 \in A'$ be the unique elements such that $f'(a'_1) = \beta(b_1)$ and $f'(a'_2) = \beta(b_2)$. Then since β and f' are R -module homomorphisms

$$f'(a'_1 + a'_2) = f'(a'_1) + f'(a'_2) = \beta(b_1) + \beta(b_2) = \beta(b_1 + b_2) ,$$

so

$$\partial(c_1 + c_2) = [a'_1 + a'_2] = [a'_1] + [a'_2] ,$$

which completes the proof that ∂ is actually an R -module homomorphism.

Finally, let us show that the sequence

$$\ker(\alpha) \xrightarrow{f|_{\ker(\alpha)}} \ker(\beta) \xrightarrow{g|_{\ker(\beta)}} \ker(\gamma) \xrightarrow{\partial} \text{coker}(\alpha) \xrightarrow{\bar{f}'} \text{coker}(\beta) \xrightarrow{\bar{g}'} \text{coker}(\gamma) \quad (1.5.2)$$

is exact. By Lemma 1.4 the sequence (1.5.2) is exact at $\ker(\beta)$ and $\text{coker}(\beta)$, so we just need to show that (1.5.2) is exact at $\ker(\gamma)$ and $\text{coker}(\alpha)$. First let us show that (1.5.2) is exact at $\ker(\gamma)$. Suppose that $c \in \text{im}(g|_{\ker(\beta)})$ so that $c = g(b)$ for some $b \in \ker(\beta)$. Then $\beta(b) = 0$, so by the definition of ∂ we have that $\partial(c) = 0$, hence $\ker(\partial) \subset \text{im}(g|_{\ker(\beta)})$.

On the other hand suppose that $c \in \ker(\partial)$. Let $b \in B$ be such that $g(b) = c$ and $a' \in A'$ be the unique element so that $f'(a') = \beta(b)$. Then by the definition of ∂ and the fact that $\partial(c) = 0$, we have that $[a'] = 0$, that is $a' \in \text{im}(\alpha)$, so there exists some element $a \in A$ so that $\alpha(a) = a'$. Thus

$$f'(a') = f'\alpha(a) = \beta f(a) .$$

But $\beta(b) = f'(a')$, so we have that $\beta(b) = \beta f(a)$. Thus $b - f(a) \in \ker(\beta)$. Since $c = g(b)$ we have that

$$c = g(b - f(a) + f(a)) = g(b - f(a)) + gf(a) = g(b - f(a)) .$$

Since $b - f(a) \in \ker(\beta)$, this shows that $c \in \text{im}(g|_{\ker(\beta)})$, as desired. Hence the sequence (1.5.2) is exact at $\ker(\gamma)$.

Now let us show that the sequence (1.5.2) is exact at $\text{coker}(\gamma)$. Suppose that $[a'] \in \text{im}(\partial)$. Then $[a'] = \partial(c)$ for some $c \in \ker(\gamma)$. We know that

$$\bar{f}'[a'] = [f'(a')] .$$

By the definition of ∂ , we have that $f'(a') = \beta(b)$, where $g(b) = c$. Hence, in $\text{coker}(\beta)$

$$\bar{f}'[a'] = [f'(a')] = [\beta(b)] = 0 ,$$

so $[a'] \in \ker(\bar{f}')$, as desired.

On the other hand, suppose that $a' \in A'$ and $\bar{f}'[a'] = 0$. This means that $f'(a') \in \text{im}(\beta)$, so $f'(a') = \beta(b)$ for some $b \in B$. Then

$$\gamma g(b) = g' \beta(b) = g' f'(a) = 0,$$

by the commutativity of (1.5.1) and the exactness of the bottom row. Thus $g(b) \in \ker(\gamma)$, and by the definition of ∂ we have that $\partial g(b) = [a']$, so $[a'] \in \text{im}(\partial)$, as desired.

The last statement, that if f is injective, then so is $f|_{\ker(\alpha)}$, and if g' is surjective, then so is $\bar{g}'$ is just an application of Lemma 1.3. $\square$

2. EXERCISES IN DIAGRAM CHASING

The following lemmas are good exercises in diagram chasing and are both standard results from homological algebra. As an exercise, try to prove these results.

2.1. Lemma (Splitting Lemma). *Let R be a commutative ring and*

$$0 \longrightarrow X \xrightarrow{\iota} Z \xrightarrow{p} Y \longrightarrow 0$$

be a short exact sequence of R -modules. The following are equivalent.

(2.1.a) *The image of ι is a direct summand of Z .*

(2.1.b) *There exists a retraction r of ι .*

(2.1.c) *There exists a section s of p .*

*In this case, we say that the sequence **splits**, and $Z \cong X \oplus Y$ in such a way that ι is the inclusion and p is the projection.*

The Snake lemma might come in handy to prove the next lemma.

2.2. Lemma (3×3 lemma). *Consider a commutative diagram*

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & M'_1 & \xrightarrow{\gamma_1} & M_1 & \xrightarrow{\rho_1} & M''_1 \longrightarrow 0 \\
 & & \downarrow \alpha' & & \downarrow \alpha & & \downarrow \alpha'' \\
 0 & \longrightarrow & M'_2 & \xrightarrow{\gamma_2} & M_2 & \xrightarrow{\rho_2} & M''_2 \longrightarrow 0 \\
 & & \downarrow \beta' & & \downarrow \beta & & \downarrow \beta'' \\
 0 & \longrightarrow & M'_3 & \xrightarrow{\gamma_3} & M_3 & \xrightarrow{\rho_3} & M''_3 \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & 0 & & 0 & & 0
 \end{array} \tag{2.2.1}$$

in $\mathbf{Mod}_R$, where all of the columns are exact and the middle row is exact. Then the first row of (2.2.1) is exact if and only if the third is.